BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 049 · Lesson 049 · Stack<T>: LIFO and Program State

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. How does List<T>.RemoveAt differ from Remove?

Q2. What is a stack's removal order?

Q3. What is an instance method?

Q4. 020

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

LIFO — Last In First Out.

Push — Add to top of stack.

Pop — Remove and return top.

Peek — View top without removing.

New learning

Observe — A stack is last-in, first-out (LIFO). Push places a value on top, Pop removes the top, and Peek reads without removing.

Explain — A stack can support undo histories and recursive call frames. The physical call stack is a related example but not this Stack<T> collection.

Apply — Always check Count or otherwise handle an empty-stack exception before Pop/Peek.

Worked example

Trace stack push RED, push BLUE, pop.

1. Trace and justify — Popped BLUE; remaining stack top RED. Most recent value leaves first.

We Do · Practise with guidance

Trace a different stack: push 2, push 5, pop, push 9. What is the top?

```
var s=new Stack<int>();
s.Push(2);s.Push(5);s.Pop();s.Push(9);
```

Trace and justify

Stack bottom-to-top [2,9], Peek returns 9; popped value was 5.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Push A, B, C; Pop once, then Peek. State both outputs and remaining stack. [6]

Question 2

Write C# to push 3,6,9 and safely pop one value. [6]

Question 3

Explain why stack suits undo history but queue suits printing jobs; include two operations. [6]

Exit Ticket · Answer before checking

Peek permanently removes the last value from a stack.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. By index versus by matching value.

Q2. LIFO.

Q3. Method called on a specific object.

Q4. Push A then B then C; pop twice; push D. Give remaining stack bottom to top.

You Do pages · Independent questions

Q1. Pop C [2], Peek B [2], remaining bottom-to-top A,B [2].

Q2. var s=new Stack<int>(); [1], Push(3),Push(6),Push(9) [3], if(s.Count>0) [1] s.Pop() [1].

Q3. Undo reverses most recent action [2] using Push/Pop [1]; printing queue serves earliest job [2] with Enqueue/Dequeue [1].

Exit Ticket · Model answer

False. Peek does not remove it.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: